

Tugas Lab. Algo Pertemuan 9

Nama : Fathurrohman

NPM : 252310040

Kelas : TI-25-PA2

Github: <https://github.com/Fathur123-TI/LabAlgoritmaSemester2.git>

Soal 1

```
1  #include <iostream>
2  using namespace std;
3
4  #define MAX 5
5
6  int stack[MAX];
7  int top = -1;
8
9  void push(int data) {
10     if (top == MAX - 1) {
11         cout << "Stack Penuh!" << endl;
12     } else {
13         top++;
14         stack[top] = data;
15     }
16 }
17
18 void tampil() {
19     if (top == -1) {
20         cout << "Stack Kosong!" << endl;
21         return;
22     }
23
24     cout << "Data stack" << endl;
25     for (int i = top; i >= 0; i--) {
26         cout << stack[i] << endl;
27     }
28 }
29
30 int main() {
31
32     push(621);
33     push(999);
34     push(201);
35     push(94);
36     push(124);
37
38     tampil();
39
40     return 0;
41 }
42
```

Output:

```
Data Stack
124
94
201
999
621
```

2.

```
1  #include <iostream>
2  using namespace std;
3
4  struct Node {
5      int data;
6      Node *prev;
7      Node *next;
8  };
9
10 class DoublyLinkedList {
11 private:
12     Node *head;
13     Node *tail;
14
15 public:
16     DoublyLinkedList() {
17         head = NULL;
18         tail = NULL;
19     }
20
21     void insert(int nilai) {
22         Node *baru = new Node;
23         baru->data = nilai;
24         baru->prev = NULL;
25         baru->next = NULL;
26
27         if (head == NULL) {
28             head = tail = baru;
29         } else {
30             baru->prev = tail;
31             tail->next = baru;
32             tail = baru;
33         }
34     }
35
36     void print() {
37         Node *temp = head;
38
39         while (temp != NULL) {
40             cout << temp->data << " ";
41             temp = temp->next;
42         }
43
44         cout << endl;
45     }
46 };
47
48 int main() {
49     DoublyLinkedList list;
50
51     list.insert(24);
52     list.insert(10);
53     list.insert(33);
54     list.insert(33);
55
56     cout << "Isi Doubly Linked List : ";
57     list.print();
58
59     return 0;
60 }
```

Output:

```
Isi Doubly Linked List : 24 10 33 33
```

```
-----
Process exited after 0.08646 seconds with return value 0
Press any key to continue . . .
```